RAJALAKSHMI ENGINEERING COLLEGE

Rakshith Y G 250701578-2025-CSE R2

Started on	Friday, 11 September 2026, 3:22 PM
State	Finished
Completed on	Friday, 11 September 2026, 3:23 PM
Time taken	37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}

int lcs(char *s1, char *s2) {
    int len1 = strlen(s1);
    int len2 = strlen(s2);

    int dp[len1 + 1][len2 + 1];

    for (int i = 0; i <= len1; i++) {
        for (int j = 0; j <= len2; j++) {
            if (i == 0 || j == 0)
                dp[i][j] = 0;
            else if (s1[i - 1] == s2[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
        }
    }
    return dp[len1][len2];
}

int main() {
    char s1[1005], s2[1005];
    if (scanf("%s %s", s1, s2) == 2) {
        printf("%d\n", lcs(s1, s2));
    }
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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